Stratification Foundations: Groups and Endowments (DRAFT)

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Abstract

This paper develops a formal framework for stratification economics that centers the role of identity and hierarchy in shaping economic outcomes. Rather than treating inequality as the result of random shocks or individual choices alone, the framework embeds group-conditioned endowments, identity-augmented preferences, stratified risks, and institutionally biased interactions directly into utility and game-theoretic structures. These features generate persistent inequality as the equilibrium outcome of stratified systems, not as an anomaly to be explained away. The framework also provides a compact parametric representation of the core stratification levers—choice breadth, information quality, constraint tightness, risk exposure, and payoff multipliers—that can be mapped into estimation-ready equations. This bridge connects theory to standard empirical methods (DiD, IV, QTE, field experiments), enabling systematic tests of stratification mechanisms and evaluation of equity-enhancing policies.

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This work also draws on foundational contributions in economics and related fields that inform the study of group-based inequality, including Kenneth J. Arrow, Gary S. Becker, Marianne Bertrand, Eduardo Bonilla-Silva, Michel Foucault, Glenn C. Loury, Karl Marx, Devah Pager, Barbara Reskin, Donald Tomaskovic-Devey, and Max Weber, among others. While these contributions are not always situated within the stratification economics tradition, they provide important conceptual and empirical foundations that this framework engages with, builds on, and in some cases challenges.

A full bibliography will be developed in future drafts.

The author used generative AI as an aid for editing, refinement, and consistency checking. All ideas originate with the author, and any AI-suggested text was carefully reviewed for accuracy and alignment with the author's voice and original text. Final content decisions and any errors remain the sole responsibility of the author.

1 Stratification Foundations: Groups and Endowments

This note establishes the first building block of the stratification framework: how power structures shape economic opportunity through the construction of social groups and the allocation of unequal endowments. It formalizes how identity conditions the distributions from which individuals draw feasible choices, information, constraints, risks, and payoffs. This foundation provides the inputs that will later determine individual decision-making (Note 2), institutional behavior (Note 3), and equilibrium outcomes (Notes 5–6).

Setup

- Population of individuals $i = 1, \ldots, N$.
- A **power structure** Υ (laws, institutions, norms) constructs social groups $G \in \mathcal{G}$ (e.g., $G \in \{\text{race}, \text{sex}, \ldots\}$) and social group identity $g \in G$ (e.g., $g \in \{\text{Black}, \text{white}, \ldots\}$).
- Each individual belongs to some identity $g \in G$ for every group $G \in \mathcal{G}$. Intersectionality means outcomes are shaped by the joint distribution of economic endowments across these multiple memberships. (e.g., Black $\in$ {race social group} $\cap$ woman $\in$ {sex social group})
- Υ endows socioeconomic conditions or **economic endowments**, to each g:[1]

$$(K_g,\ I_g,\ \Lambda_g,\ P_g,\ \Pi_g)\ =\ \Phi\left(\Upsilon, g \mid G\right)$$

  where:

  - $K_g = K(\kappa_g) \equiv$ **choice set**, with breadth determined by κ_g (e.g., range of schooling, jobs, or neighborhoods legally or practically accessible),
  - $I_g = I(\iota_g) \equiv$ **information set**, with quality/clarity determined by ι_g (e.g., advising, hiring transparency, wage posting, networks),
  - $\Lambda_g = \Lambda(\lambda_g) \equiv$ **constraint set**, with tightness determined by λ_g (e.g., credit ceilings, budget constraints, exclusion rules),
  - $P_g = P(\rho_g) \equiv$ **risk environment**, with exposure summarized by ρ_g (e.g., income volatility, health/policing risk, deviance penalty risk),
  - $\Pi_g = \Pi(\pi_g) \equiv$ **payoff schedule**, with multiplier π_g (e.g., returns to education, wage offers, penalties).

Intuition: Υ stratifies groups by endowing more favorable $(\kappa, \iota, \lambda, \rho, \pi)$ to dominant identities and less favorable endowments to subaltern, or lower status, identities. In other words: society's rules distribute opportunity differently depending on your group identity(ies).

Individual Variation Within Identity-Level Endowments

To capture that stratification operates "on average" while allowing within-identity variation and cross-identity overlap, let each individual i draw an *individual economic endowment* vector

$$\omega_i\ =\ (\kappa_i,\ \iota_i,\ \lambda_i,\ \rho_i,\ \pi_i)$$

[1] Given a social group G, the power structure applies mapping function Φ to unequally distribute endowments across all group identities $g \in G$.

from a identity-specific distribution D_g centered on the identity endowments $\Omega_g = (\kappa_g, \iota_g, \lambda_g, \rho_g, \pi_g)$:

$$\omega_i \;\sim\; D_g\left(\mu = \Omega_g,\, \Sigma_g\right)\,,$$

where Σ_g is the variance-covariance matrix capturing within-identity dispersion. Under stratification, D_g first order stochastically dominates $D_{g'}$ when g is the dominant identity and g' the subaltern identity.

Given ζ_i, an individual-specific noise multiplier vector, a convenient positive-support parameterization of ω_i is elementwise multiplicative noise:[2]

$$\omega_i \;=\; \Omega_g \;\odot\; \zeta_i, \qquad \zeta_i > 0, \quad \mathbb{E}\left[\zeta_i \mid g_i\right] = \mathbf{1}.$$

Thus, identity means are preserved, with dispersion Σ_g capturing within-identity variation, and allowing overlap across identities. Accordingly, economic endowments are *individual-specific but identity-conditioned*:

$$\begin{aligned}
K_i &= K\left(\kappa_i\right) \equiv i\text{'s } \textbf{choice set},\\
I_i &= I\left(\iota_i\right) \equiv i\text{'s } \textbf{information set},\\
\Lambda_i &= \Lambda\left(\lambda_i\right) \equiv i\text{'s } \textbf{constraint set},\\
P_i &= P\left(\rho_i\right) \equiv i\text{'s } \textbf{risk environment},\\
\Pi_i &= \Pi\left(\pi_i\right) \equiv i\text{'s } \textbf{payoff schedule}.
\end{aligned}$$

Intuition: Stratification sets up unequal, identity-specific economic endowment lotteries from which individuals (given identity membership) draw. While there is variation within the lotteries (captured by Σ_g), and some overlap across lotteries, individuals draw from distributions centered on their group identity's average, with dominant identities systematically advantaged relative to subaltern identities.

Benchmark foundation (individualist view). Taking the individualist view, assume every individual draws from a single population economic endowment pool. Under this benchmark, all individuals draw from a common endowment distribution. Persistent inequality between identities in outcomes is therefore typically interpreted as arising from differences in ability, effort, or culture:

Population economic endowments:

$$\left(K^{\text{pop}},\; I^{\text{pop}},\; \Lambda^{\text{pop}},\; P^{\text{pop}},\; \Pi^{\text{pop}}\right) \;=\; \Phi^{\text{pop}}$$

Individual variation:
Let individual i draw an individual economic endowment vector

$$\omega_i^{\text{pop}} \;=\; \left(\kappa_i^{\text{pop}},\; \iota_i^{\text{pop}},\; \lambda_i^{\text{pop}},\; \rho_i^{\text{pop}},\; \pi_i^{\text{pop}}\right)$$

from distribution D^{pop} centered on population endowments $\Omega^{\text{pop}} = (\kappa^{\text{pop}}, \iota^{\text{pop}}, \lambda^{\text{pop}}, \rho^{\text{pop}}, \pi^{\text{pop}})$:

$$\omega_i^{\text{pop}} \;\sim\; D^{\text{pop}}\left(\mu = \Omega^{\text{pop}},\, \Sigma^{\text{pop}}\right)\,,$$

by applying individual-specific noise multiplier vector:

$$\omega_i^{\text{pop}} \;=\; \Omega^{\text{pop}} \;\odot\; \zeta_i^{\text{pop}}, \qquad \zeta_i^{\text{pop}} > 0, \quad \mathbb{E}\left[\zeta_i^{\text{pop}}\right] = \mathbf{1}.$$

[2] Alternatively, one may apply an additive noise vector: $\omega_i = \Omega_g + \varepsilon_i$ with truncated support for nonnegativity. Multiplicative noise implies a log-normal D_g, which is often simpler for estimation and preserves positivity by construction.

Compact Intuition: Groups and Endowments

Setup/Inputs: Power structures construct social groups and hierarchies, define identities, and allocate unequal endowment pools (choices, information, constraints, risks, payoffs) across identities. Individuals then draw from these identity-conditioned pools to determine their own endowments.

Choices/Interactions: These endowments set the opportunity frontier for each identity, shaping how individuals will be able to act and how institutions or others will interact with them.

Outcomes/Solution: Because dominant identities receive systematically better endowments, their members achieve better average outcomes, while subaltern identities face worse outcomes. Inequality is embedded from the start, before preferences or behavior enter the picture.

Table 1: Comparing Perspectives on Foundations: Individualist vs. Structuralist

Dimension	**Individualist Perspective**	**Structuralist Perspective**
Power Structures	Institutions are identity-neutral or irrelevant background.	Power structures Υ actively construct groups and hierarchies.
Group Identities	Exogenous or irrelevant.	Socially constructed, intersectional, and stratified.
Endowments/Inputs	All individuals draw from a common endowment pool.	Endowments are identity-conditioned: Ω_g sets group averages, $\omega_i \sim D_g$ allows within-identity variation.
Decision Rules	Inequality reflects random variation, ability, or culture.	Stratification ensures dominant identities stochastically dominate subaltern identities.

Bringing It Together

This note establishes the "rules of the game." Power structures (Υ) construct and stratify social group identities, then assign unequal economic endowments—choice sets, information, constraints, risks, and payoffs—to each identity. E.g., two students with equal ability, but one faces tighter credit and weaker schools because of race. These endowments define identity-level averages while allowing within-identity variation and cross-identity overlap through the individual "endowment lottery." As a result, dominant groups systematically draw from more favorable distributions than subaltern identities. E.g., two equally skilled workers, but one faces tighter credit, fewer networks, and lower wage offers because of identity. This foundation sets the stage for how identity and preferences (Note 2) and subsequent behavior are conditioned by stratification.